

THE BFGS METHOD

Lecture 19: Presented by Jan van Vuuren



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The BFGS update formula (p. 93)

It can be shown that a rank-two update rule $\mathbf{X}^{k+1} = \mathbf{X}^k + \mathbf{\Delta}^k$ of a sequence $\mathbf{X}^0, \mathbf{X}^1, \mathbf{X}^2, \dots$ of matrix estimates preserves symmetry of the estimates if and only if the iterative correction term assumes the form

$$\mathbf{\Delta}^k = a^k \mathbf{q}^k (\mathbf{q}^k)^T + b^k \mathbf{r}^k (\mathbf{r}^k)^T$$

for some n -vectors $\mathbf{q}^k$ and $\mathbf{r}^k$, where a^k and b^k are scalars

The four British/American mathematicians

- Charles Broyden (1933–2011)
- Roger Fletcher (1939–2016)
- Donald Goldfarb (1941–)
- David F. Shanno (1938–2019)

studied this update rule independently in 1970. They showed that by taking $\mathbf{q}^k = \mathbf{y}^k$ and $\mathbf{r}^k = \mathbf{B}^k \mathbf{s}^k$ in the update rule above, the secant condition $\mathbf{B}^{k+1} \mathbf{s}^k = \mathbf{y}^k$ is satisfied when estimating the Hessian matrix $\nabla^2 f(\mathbf{x}^k)$, if

$$a^k = \frac{1}{(\mathbf{y}^k)^T \mathbf{s}^k} \quad \text{and} \quad b^k = \frac{-1}{(\mathbf{s}^k)^T (\mathbf{B}^k)^T \mathbf{s}^k}$$

The BFGS update formula (p. 93)

This leads to the Hessian matrix approximation update rule

$$\begin{aligned} B^{k+1} &= B^k + \Delta^k \\ &= B^k + a^k \mathbf{q}^k (\mathbf{q}^k)^T + b^k \mathbf{r}^k (\mathbf{r}^k)^T \\ &= B^k + \left(\frac{1}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) \mathbf{y}^k (\mathbf{y}^k)^T + \left(\frac{-1}{(\mathbf{s}^k)^T (B^k)^T \mathbf{s}^k} \right) (B^k \mathbf{s}^k) (B^k \mathbf{s}^k)^T \\ &= B^k + \frac{\mathbf{y}^k (\mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} - \frac{B^k \mathbf{s}^k (B^k \mathbf{s}^k)^T}{(\mathbf{s}^k)^T B^k \mathbf{s}^k} \end{aligned}$$

Applying the Sherman-Morrison formula to this expression yields the alternative update rule

$$\mathbf{H}^{k+1} = \left(\mathbf{I} - \frac{\mathbf{s}^k (\mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) \mathbf{H}^k \left(\mathbf{I} - \frac{\mathbf{y}^k (\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) + \frac{\mathbf{s}^k (\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k}$$

The BFGS method in pseudo-code (p. 94)

Algorithm 3.13 (The Broyden-Fletcher-Goldfarb-Shanno method)

Input: An instance of a multivariate differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\mathbf{x}^0$ as to an approximate local optimum of f , an initial Hessian estimate $\mathbf{B}^0$ or an initial inverse Hessian estimate $\mathbf{H}^0$, as well as a search termination criterion

Output: An approximation of a local optimum of f close to $\mathbf{x}^0$ satisfying the termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search termination criterion is met, then output $\mathbf{x}^k$ and stop

Step 3 Compute the search direction $\mathbf{p}^k$ by Gauss-Jordan elimination to solve the system of linear equations $\mathbf{B}^k \mathbf{p}^k = -\nabla f(\mathbf{x}^k)$, or compute it directly as $\mathbf{p}^k = -\mathbf{H}^k \nabla f(\mathbf{x}^k)$

Step 4 Compute an appropriate step length α^k in the search direction $\mathbf{p}^k$ by performing an exact line search or a backtracking Armijo-Goldstein line search in respect of the function $f(\mathbf{x}^k + \alpha^k \mathbf{p}^k)$

Step 5 Compute the next search estimate as $\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha^k \mathbf{p}^k$

Step 6 $\mathbf{s}^k \leftarrow \mathbf{x}^{k+1} - \mathbf{x}^k$

Step 7 $\mathbf{y}^k \leftarrow \nabla f(\mathbf{x}^{k+1}) - \nabla f(\mathbf{x}^k)$

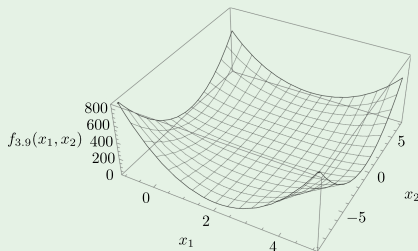
Step 8 $\mathbf{B}^{k+1} \leftarrow \mathbf{B}^k + \frac{\mathbf{y}^k (\mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} - \frac{\mathbf{B}^k \mathbf{s}^k (\mathbf{B}^k \mathbf{s}^k)^T}{(\mathbf{s}^k)^T \mathbf{B}^k \mathbf{s}^k}$ or

$$\mathbf{H}^{k+1} \leftarrow \left(\mathbf{I} - \frac{\mathbf{s}^k (\mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) \mathbf{H}^k \left(\mathbf{I} - \frac{\mathbf{y}^k (\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) + \frac{\mathbf{s}^k (\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k}$$

Step 9 $k \leftarrow k + 1$ and return to Step 2

Example 3.17 (pp. 94–96)

$$f_{3.9}(x_1, x_2) = 17 - 32x_1 + 24x_1^2 - 8x_1^3 + x_1^4 + 2x_2 + 5x_2^2 - 4x_1x_2^2 + x_1^2x_2^2$$



We apply Algorithm 3.13 (equipped with a Armijo-Goldstein line search with parameters $a = 0.4$ and $b = 0.9$) to minimise $f_{3.9}$, starting out from $\mathbf{x}^0 = [-1 \ 1]^T$

We are given an initial symmetric, positive definite Hessian estimate

$$\mathbf{B}^0 = \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$$

We work accurately to six decimal places and terminate the iterative process once the 2-norm $\|\nabla f_{3.9}(\mathbf{x}^k)\|_2$ drops below 0.05 for the first time.

Example 3.17 (pp. 94–96)

Our first step is to compute the gradient vector of $f_{3.9}$ as

$$\nabla f_{3.9}(x_1, x_2) = \begin{bmatrix} 2(x_1 - 2)(8 - 8x_1 + 2x_1^2 + x_2^2) \\ 2 + 2(5 - 4x_1 + x_1^2)x_2 \end{bmatrix}$$

During Iteration $k = 0$, the gradient vector is $\nabla f_{3.9}(\mathbf{x}^0) = [-114 \ 22]^T$

We compute the search direction by solving the system

$$\begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} p_1^0 \\ p_2^0 \end{bmatrix} = \begin{bmatrix} 114 \\ -22 \end{bmatrix}$$

The first search direction $\mathbf{p}^0 = [\frac{46}{5} \ \frac{37}{10}]^T = [9.2 \ 3.7]^T$ is obtained

A backtracking Armijo-Goldstein line search with parameters $a = 0.4$ and $b = 0.9$ yields the step length $\alpha^0 = 0.205891$

As a result, $\mathbf{s}^0 = \alpha^0 \mathbf{p}^0 \approx [1.894198 \ 0.761797]^T$

Therefore, $\mathbf{x}^1 = \mathbf{x}^0 + \alpha^0 \mathbf{p}^0 \approx [0.894198 \ 1.761797]^T$

Moreover, $\mathbf{y}^0 = \nabla f_{3.9}(\mathbf{x}^1) - \nabla f_{3.9}(\mathbf{x}^0) \approx [101.726656 \ -12.167764]^T$

Example 3.17 (pp. 94–96)

The output of the BFGS method coincides with those of Broyden's method and the SR1 method during the first iteration ... the only difference comes in after the first Hessian update

In anticipation of the update rule

$$\mathbf{B}^1 = \mathbf{B}^0 + \frac{\mathbf{y}^0(\mathbf{y}^0)^T}{(\mathbf{y}^0)^T \mathbf{s}^0} - \frac{\mathbf{B}^0 \mathbf{s}^0 (\mathbf{B}^0 \mathbf{s}^0)^T}{(\mathbf{s}^0)^T \mathbf{B}^0 \mathbf{s}^0},$$

the following intermediate calculations are first performed during iteration 1:

$$\begin{aligned}\mathbf{B}^0 \mathbf{s}^0 &= \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} 1.894\,198 \\ 0.761\,797 \end{bmatrix} = \begin{bmatrix} 23.471\,584 \\ -4.529\,604 \end{bmatrix} \\ (\mathbf{y}^0)^T \mathbf{s}^0 &= (101.726\,656)(1.894\,198) + (-12.167\,764)(0.761\,797) \\ &= 183.421\,062, \\ (\mathbf{s}^0)^T \mathbf{B}^0 \mathbf{s}^0 &= (1.894\,198)(23.471\,584) + (0.761\,797)(-4.529\,604) \\ &= 41.009\,189.\end{aligned}$$

Example 3.17 (pp. 94–96)

The Hessian estimate is consequently updated to

$$\begin{aligned} \mathbf{B}^1 &= \mathbf{B}^0 + \frac{\mathbf{y}^0(\mathbf{y}^0)^T}{(\mathbf{y}^0)^T \mathbf{s}^0} - \frac{\mathbf{B}^0 \mathbf{s}^0 (\mathbf{B}^0 \mathbf{s}^0)^T}{(\mathbf{s}^0)^T \mathbf{B}^0 \mathbf{s}^0} \\ &= \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} + \frac{1}{183.421\,062} \begin{bmatrix} 101.726\,656 \\ -12.167\,764 \end{bmatrix} [101.726\,656 \quad -12.167\,764] \\ &\quad - \frac{1}{41.009\,189} \begin{bmatrix} 23.471\,584 \\ -4.529\,604 \end{bmatrix} [23.471\,584 \quad -4.529\,604] \\ &\approx \begin{bmatrix} 56.984\,385 & -8.115\,813 \\ -8.115\,813 & 4.306\,873 \end{bmatrix} \end{aligned}$$

Since

$$\|\nabla f_{3.9}(\mathbf{x}^0)\|_2 \approx 116.103\,402 > 0.05,$$

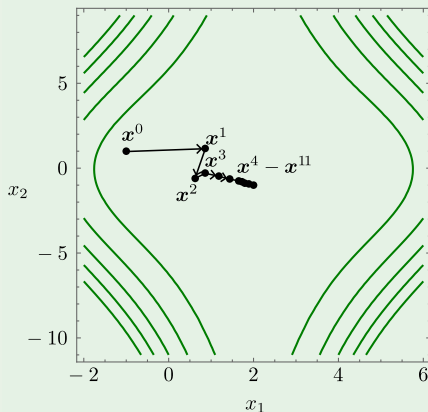
Algorithm 3.12 does not terminate immediately after the first iteration ...

Example 3.17 (pp. 94–96)

k	$\mathbf{x}^k$	$\nabla f_{3.9}(\mathbf{x}^k)$	$\mathbf{B}^k$	$\mathbf{p}^k$	α^k	$\mathbf{s}^k$	$\mathbf{y}^k$	$\ \nabla f_{3.9}(\mathbf{x}^k)\ _2$
0	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -114 \\ 22 \end{bmatrix}$	$\begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$	$\begin{bmatrix} 9.2 \\ 3.7 \end{bmatrix}$	0.206	$\begin{bmatrix} 1.894 \\ 0.762 \end{bmatrix}$	$\begin{bmatrix} 101.727 \\ -12.168 \end{bmatrix}$	116.103
1	$\begin{bmatrix} 0.894 \\ 1.762 \end{bmatrix}$	$\begin{bmatrix} -12.273 \\ 9.832 \end{bmatrix}$	$\begin{bmatrix} 56.984 & -8.156 \\ -8.156 & 4.307 \end{bmatrix}$	$\begin{bmatrix} -0.153 \\ -2.572 \end{bmatrix}$	0.9	$\begin{bmatrix} -0.137 \\ -2.315 \end{bmatrix}$	$\begin{bmatrix} 3.825 \\ -10.649 \end{bmatrix}$	15.726
2	$\begin{bmatrix} 0.757 \\ -0.553 \end{bmatrix}$	$\begin{bmatrix} -8.448 \\ -0.817 \end{bmatrix}$	$\begin{bmatrix} 51.158 & -4.691 \\ -4.691 & 4.879 \end{bmatrix}$	$\begin{bmatrix} 0.198 \\ 0.358 \end{bmatrix}$	1	$\begin{bmatrix} 0.198 \\ 0.358 \end{bmatrix}$	$\begin{bmatrix} 3.799 \\ 1.998 \end{bmatrix}$	8.488
3	$\begin{bmatrix} 0.955 \\ -0.195 \end{bmatrix}$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
9	$\begin{bmatrix} 1.971 \\ -1.049 \end{bmatrix}$	$\begin{bmatrix} -0.063 \\ -0.099 \end{bmatrix}$	$\begin{bmatrix} 2.124 & 0.203 \\ 0.203 & 1.976 \end{bmatrix}$	$\begin{bmatrix} 0.025 \\ 0.048 \end{bmatrix}$	1	$\begin{bmatrix} 0.025 \\ 0.048 \end{bmatrix}$	$\begin{bmatrix} 0.056 \\ 0.097 \end{bmatrix}$	0.118
10	$\begin{bmatrix} 1.997 \\ -1.001 \end{bmatrix}$	$\begin{bmatrix} -0.007 \\ -0.002 \end{bmatrix}$	$\begin{bmatrix} 2.016 & 0.114 \\ 0.114 & 1.976 \end{bmatrix}$	$\begin{bmatrix} 0.003 \\ 0.001 \end{bmatrix}$	1	$\begin{bmatrix} 0.003 \\ 0.001 \end{bmatrix}$	$\begin{bmatrix} 0.007 \\ 0.002 \end{bmatrix}$	0.007
11	$\begin{bmatrix} 2.000 \\ -1.000 \end{bmatrix}$	—	—	—	—	—	—	—

Example 3.17 (pp. 94–96)

The BFGS method returns $\mathbf{x}^{11} = [1.999\,927 \quad -1.000\,170]^T$ as approximate minimum of $f_{3.9}$; Compare this to the true minimum is $\mathbf{x}^* = [2 \quad -1]^T$



Advantages of the BFGS method (p. 97)

An advantage of the BFGS update rule is that it preserves both symmetry and positive (or negative) definiteness of the Hessian estimates by design

This makes it another popular method among practitioners.

Broyden, Fletcher, Goldfarb, Shanno

